

Project 02 — CSS & Visual Presentation

Designing Modern Interfaces

Summary: This document defines the subject matter for Project 02: Flexbox layout mechanics, CSS positioning contexts, pure CSS geometry, color models and alpha transparency, box model calculations, keyframe animations, and interactive transitions.

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Chapter I: Instructions

- These exercises are carefully arranged in order of difficulty, from easiest to hardest. We will not consider a successfully completed harder exercise if an easier one is not fully functional.
- Ensure that you have the appropriate permissions and correct directory structure for all your files.
- You must follow the exact submission procedures and naming conventions for every exercise.
- Your exercises will be checked and graded by your fellow classmates during peer evaluations.
- Additionally, your exercises may be validated by automated evaluation scripts. The evaluation is rigorous and strict. To avoid unpleasant surprises, be as thorough as possible.
- You must not leave any extraneous files in your directories other than those explicitly specified in the subject.
- Have a question? Ask the peer on your right. If not, try the peer on your left.
- Your reference guides are official documentation (MDN, man, Git docs) and verified technical resources.
- Examine the examples and edge cases carefully. They contain critical details.

— *Did you check with your right-side neighbor?* —

Chapter II: AI Instructions

The Web Piscine is foundational. Your primary mission is to build solid problem-solving reflexes, genuine technical autonomy, and collaborative peer learning.

In the modern tech ecosystem, generative AI shortcuts are tempting. However, shortcuts prevent the formation of neural connections required for deep engineering intuition.

Key Rules on AI Usage:

- Apply your own reasoning to each exercise before consulting external AI tools.
- Never ask an AI to generate solutions directly for you. You must write every line of code yourself.
- During peer evaluation and exams, you will have to explain every architectural decision and command. If you cannot explain your code, you will receive zero points.
- Good practice: You are confused about a concept like Flexbox or Git staging. You discuss it with your peer for 10 minutes, sketch it on paper, and it clicks.
- Bad practice: Copying code without understanding. You fail the peer defense because you cannot answer conceptual questions.

— *Learning is not knowing the answer; it is developing the ability to find it.* —

Chapter III: Foreword

If HTML is the bones of a building, CSS is the architecture, the paint, and the interior design.

CSS (Cascading Style Sheets) turns plain, unstyled documents into visually engaging digital experiences. Master layout, positioning, shapes, and color to command the visual canvas.

— Style separates an unformatted document from a polished web app. —

Chapter IV — Exercise 00: Flexbox Layout

Directory: ex00/

Files to Submit: index.html, style.css

Authorized: display: flex, justify-content, align-items, gap

The 1-dimensional layout engine.

- Create a navigation bar and a responsive card grid using CSS Flexbox.
- Configure .nav-container with display: flex, justify-content: space-between, and align-items: center.
- Configure .card-container with flex-direction: row, flex-wrap: wrap, and gap: 20px.

! Do not use CSS float tricks or HTML layout tables.

— Understand your main axis and cross axis. —

Chapter V — Exercise 01: Positioning

Directory: ex01/

Files to Submit: index.html, style.css

Authorized: static, relative, absolute, fixed, z-index

Commanding element placement.

- Construct a page featuring:
 1. A pinned navigation bar with `position: fixed; top: 0;`.
 2. A card container with `position: relative;`.
 3. An absolute notification badge anchored inside the card corner.
 4. A fixed back-to-top button at the bottom-right of the viewport.

! Ensure the absolute badge anchors inside its card, not the entire page.

— *Absolute elements seek the nearest positioned ancestor.* —

Chapter VI — Exercise 02: CSS Shapes

Directory: ex02/

Files to Submit: index.html, style.css

Authorized: width, height, border, border-radius

Drawing geometry with pure CSS.

- In index.html, create empty div elements and style them into 5 distinct shapes:

1. Square, 2. Rectangle, 3. Perfect Circle (border-radius: 50%), 4. Capsule Pill, 5. CSS Triangle using border tricks.

! No images or SVG files are allowed; draw purely with CSS properties.

— Geometry in the browser is made of borders and curves. —

Chapter VII — Exercise 03: Colors & Opacity

Directory: ex03/

Files to Submit: index.html, style.css

Authorized: Named, HEX, RGB, RGBA, opacity

Mastering the visual spectrum.

- Build a color swatch palette demonstrating named colors, 6-digit HEX codes, RGB, and RGBA with alpha transparency.
- Compare an element using opacity: 0.5 with an element using background-color: rgba(..., 0.5).
- Demonstrate why opacity fades children while RGBA isolates background transparency.

— Opacity cascades to all children; RGBA affects only its property. —

Chapter VIII — Exercise 04: The Box Model

Directory: ex04/

Files to Submit: index.html, style.css

Authorized: content-box vs border-box, padding, margin, border

The fundamental math of web layout.

- Create two comparison boxes illustrating box-sizing: content-box versus box-sizing: border-box.
- Demonstrate how border-box absorbs padding and borders inside the declared width.
- Implement the universal reset (*, *::before, *::after) on the page.

— *Inspect the box model diagram in DevTools.* —

Chapter IX — Bonus: Animations & Transitions

Directory: ex05-bonus/, ex06-bonus/

Files to Submit: index.html, style.css

Authorized: @keyframes, transition, transform

Bringing stylesheets to life.

- ex05-bonus: Build a smooth, looping loader animation using @keyframes, animation-duration, and animation-iteration-count: infinite.
- ex06-bonus: Build interactive hover micro-interactions on cards and buttons using transition and transform: translateY(-8px) scale(1.05).

— *Micro-interactions make static interfaces feel responsive and alive.* —

Chapter XIV: Submission & Evaluation

- Submit your assignment to your Git repository as instructed. Only the work pushed to your repository branch will be evaluated during defense.
- Double-check your directory structure, file permissions, and exact file names before final submission.

! You must submit only the files explicitly required by the project instructions.

- Prepare to demonstrate your code live, explain your logic line-by-line, and answer questions about core concepts.

— RTFM and verify your work before peer defense! —